Exact values of the Bougard–Joret function in its open second regime up to order 22

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github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

Let $f(n,\alpha,k)$ be the minimum number of edges of a k-connected graph of order n with independence number α. Bougard and Joret conjectured $f(n,\alpha,k) = t(n,\alpha) + \lceil k\alpha/2 \rceil$ when $n > k\alpha$, where $t(n,\alpha)$ is the size of the balanced union of α cliques. Das and Gupta disproved the other line of the conjecture and left this second regime open. Bougard and Joret proved the second line only when $\lfloor n/\alpha \rfloor \geq \lceil k\alpha/2 \rceil - \alpha + 2$ (for $\alpha \geq 3$). We compute f exactly at every triple outside that range with $\alpha \geq 3$, $k \geq 3$ and $n \leq 22$: 39 triples. At every one of them the value equals the conjectured one, by complete enumeration of every candidate graph below it (6,217,198,475 candidate graphs in all). The enumeration uses nauty with two pruning rules that are proved sound, and an exact filter checked against an independent implementation. The data support the second line of the conjecture in its smallest open cases; they do not prove it in general.

1 Problem and prior work

All graphs are finite and simple. Write $\mathrm{e}(G)$ for the number of edges, $\alpha(G)$ for the independence number and $\kappa(G)$ for the vertex connectivity. For $n \geq 2\alpha$, $n \geq \alpha + k$, $\alpha \geq 2$ and $k \geq 3$, Bougard and Joret conjectured [1, Conjecture 1]

$$f(n,\alpha,k) = \begin{cases} \lceil nk/2 \rceil, & n \leq k\alpha, \\ t(n,\alpha) + \lceil k\alpha/2 \rceil, & n > k\alpha. \end{cases}$$

The upper bound in the second regime holds by their construction: the balanced union of α cliques $T(n,\alpha)$ becomes k-connected after adding $\lceil k\alpha/2 \rceil$ edges without raising the independence number [1, Section 6]. Das and Gupta determined the boundary $n = \alpha + k$, which disproves the first line (the smallest counterexample is $f(7,3,4) = 15$), and left the second regime open [2]. The first interior shell $n = \alpha + k + 1$ [3] and the value $f(2d+3, 2, d+2) = d^2 + 4d + 2$ [4] lie in the first regime.

What is proved in the second regime. Bougard and Joret's Proposition 8 states that a 2-connected graph with $n \geq 3\alpha \geq 9$ and at most $t(n,\alpha) + \lfloor n/\alpha \rfloor + \alpha - 3$ edges has a vertex partition into α cliques [1]. A k-connected graph with such a partition has at least $t(n,\alpha) + \lceil k\alpha/2 \rceil$ edges (each clique sends at least k edges out of it). Hence the second line holds whenever $\lceil k\alpha/2 \rceil - 1 \leq \lfloor n/\alpha \rfloor + \alpha - 3$, that is

$$\lfloor n/\alpha \rfloor \;\geq\; \lceil k\alpha/2 \rceil - \alpha + 2 \qquad (\alpha \geq 3), \tag{1}$$

and also for $\alpha = 2$, $n \geq 2k$ [1]. The *open window* is the set of second-regime triples with $\alpha \geq 3$ violating (1). Its smallest members are $(10, 3, 3)$ and $(11, 3, 3)$.

2 Method

For a triple (n, α, k) with conjectured value $c = t(n,\alpha) + \lceil k\alpha/2 \rceil$ we decide, for every edge count $E < c$ that can occur, whether some graph on n vertices with E edges has independence number exactly α and connectivity at least k. If none does, $f(n,\alpha,k) \geq c$, and with the construction, $f(n,\alpha,k) = c$. Two classical facts bound the edge counts: minimum degree at least k gives $E \geq \lceil nk/2 \rceil$, and Turán's theorem applied to the complement gives $E \geq t(n,\alpha)$ for independence number at most α [5]. So only $\lceil k\alpha/2 \rceil$ edge counts per triple need to be searched.

Every graph with n vertices, E edges and minimum degree at least k is generated up to isomorphism by `geng` from nauty 2.8.9 [6], split into disjoint parts. Each graph is tested by an exact filter: the independence number by branch and bound on bit sets, and connectivity at least k by deleting every vertex set of size less than k. Two refinements were added during the work; each is proved sound and was gated before use.

Lemma 1 (early stop)**.** *Rejecting a graph as soon as the search finds an independent set of size $\alpha + 1$ leaves the decision "$\alpha(G) = \alpha$" unchanged.*

Proof. Such a graph has $\alpha(G) \geq \alpha + 1$ and is rejected by the full computation as well; otherwise the search runs to the end and computes $\alpha(G)$ exactly. □

Lemma 2 (independence prune)**.** *During generation, a partial graph that already contains an independent set of size $\alpha + 1$ can be discarded with all its extensions.*

Proof. `geng` extends a graph by adding vertices only, so the partial graph is an induced subgraph of each of its completions. The independence number of an induced subgraph never exceeds that of the graph, so every completion has independence number at least $\alpha + 1$ and is rejected. □

With Lemma 2, the $(17, 5, 3)$ search at $E = 27$ dropped from 1,040 seconds (35,329,226 candidates) to under a second (482 candidates), with the same answer.

3 Validation

- The filter agrees with an independent implementation (networkx) on all 12,346 graphs on 8 vertices, for every $\alpha \leq 6$ and $k \leq 5$, before and after Lemma 1.
- Known values are reproduced: $f(7,3,4) = 15$ [2], $f(6,2,3) = 9$, $f(6,2,4) = 12$, and the second-regime value $f(12,3,3) = 23$ inside the proved range (1).

- The generator with Lemma 2 reproduces the numbers of qualifying graphs found without it at twelve known cases, from $(7, 3, 4)$ to $(17, 5, 3)$.
- Connected-graph counts of `geng` agree with OEIS A001349 where checked, and the 4,207 graphs at $(16, 5, 3)$, $E = 24$ are the cubic graphs on 16 vertices (OEIS A005638).

4 Results

Theorem 3 (MV)**.** *At each of the 39 triples of the open window with $n \leq 22$ (Table 1), no graph with fewer edges than the conjectured value has independence number α and connectivity at least k; hence $f(n, \alpha, k) = t(n, \alpha) + \lceil k\alpha/2 \rceil$ at every one of them.*

5 Scope

The open window is infinite; these computations cover its members of order at most 22 and nothing beyond. The results are exact finite statements, verified by machine; the general second line of the conjecture remains open. The extremal graphs found at the conjectured values are listed in the computational record [7].

Data availability. Code, records and the lists of extremal graphs are in the repository [7], experiments EXP-005, EXP-006 and EXP-007 under `problems/combinatorics/bougard-joret/experiments/`.

Declaration of generative AI use. Generative AI tools were used for literature triage, derivation exploration, code drafting and internal review of the text. The author reviewed and verified all content and takes full responsibility for the work.

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(n, α, k)	$t(n, \alpha)$	f	edges searched	graphs below	extremal graphs
$(10, 3, 3)$	12	17	15–16	383	15
$(11, 3, 3)$	15	20	17–19	55,812	6
$(13, 3, 4)$	22	28	26–27	1,095,032	1
$(13, 4, 3)$	15	21	20–20	1,999	1
$(14, 3, 4)$	26	32	28–31	5,268,961,461	construction
$(14, 4, 3)$	18	24	21–23	2,147,042	1
$(15, 4, 3)$	21	27	23–26	904,211,641	1
$(16, 3, 5)$	35	43	40–42	134	3
$(16, 5, 3)$	18	26	24–25	932,175	18
$(17, 3, 5)$	40	48	43–47	2,120	3
$(17, 4, 4)$	28	36	34–35	99	2
$(17, 5, 3)$	21	29	26–28	35,627,553	34
$(18, 3, 5)$	45	53	45–52	6,568	2
$(18, 4, 4)$	32	40	36–39	4,086	4
$(18, 5, 3)$	24	32	27–31	71,021	35
$(19, 3, 5)$	51	59	51–58	17,920	4
$(19, 3, 6)$	51	60	57–59	83	1
$(19, 4, 4)$	36	44	38–43	24,673	2
$(19, 5, 3)$	27	35	29–34	149,597	18
$(19, 6, 3)$	21	30	29–29	295	2
$(20, 3, 5)$	57	65	57–64	21,372	4
$(20, 3, 6)$	57	66	60–65	4,046	1
$(20, 4, 4)$	40	48	40–47	41,808	2
$(20, 5, 3)$	30	38	30–37	132,654	5
$(20, 6, 3)$	24	33	30–32	23,316	5
$(21, 3, 6)$	63	72	63–71	29,091	1
$(21, 4, 4)$	45	53	45–52	141,850	2
$(21, 4, 5)$	45	55	53–54	420	1
$(21, 5, 3)$	34	42	34–41	471,911	17
$(21, 5, 4)$	34	44	42–43	1,298	7
$(21, 6, 3)$	27	36	32–35	303,615	5
$(22, 3, 6)$	70	79	70–78	83,931	1
$(22, 3, 7)$	70	81	77–80	782	construction
$(22, 4, 4)$	50	58	50–57	222,379	4
$(22, 4, 5)$	50	60	55–59	47,945	construction
$(22, 5, 3)$	38	46	38–45	874,494	31
$(22, 5, 4)$	38	48	44–47	101,779	construction
$(22, 6, 3)$	30	39	33–38	1,365,978	5
$(22, 7, 3)$	24	35	33–34	20,112	construction

Table 1: The open window up to order 22. Column f: the exact value (the conjectured value when the search below it is complete). "Edges searched": the edge counts below the conjectured value that can occur. "Graphs below": the candidate graphs generated at those edge counts (with the generator in use; the counts after Lemma 2 are smaller). "Extremal graphs": the qualifying graphs at the conjectured value when that edge count was also enumerated; otherwise the attaining construction [1].